

Day 0E Masterclass

Calculus for Machine Learning

Use the arrow keys. Mark a segment reviewed after you can re-derive its result.

Segment 1 • What a slope is, and why a curve does not have one

A derivative is the slope at one exact point.

MHP cable length and cost form a line before we study curves.
Pause, type, break, rebuild.

Segment 2 • The limit definition, derived not asserted

A derivative is a limit of nearby average slopes.

At a terrain index of 3, the numerical slope approaches 6.
Pause, type, break, rebuild.

Segment 3 • The rules you will actually use

Power, constant-multiple, and sum rules are compressed limit calculations.

$$n$$
$$n^{-1}$$

A cost curve with cable length and terrain index is differentiated term by term.

Pause, type, break, rebuild.

Segment 4 • Product, quotient, and chain rules

Nested functions pass sensitivity through each layer.

The squared MHP cost error gives the same gradient by two roads.

Pause, type, break, rebuild.

Segment 5 • Exponential and logarithm

Exponential growth reproduces itself; logarithms undo it.

x

x

Log-transformed million-PKR costs change the derivative scale.
Pause, type, break, rebuild.

Segment 6 • Sigmoid, tanh, and ReLU

Activation derivatives explain saturation and vanishing gradients.

A risk score for an MHP warning can become insensitive in a saturated tail.

Pause, type, break, rebuild.

Segment 7 • Second derivatives, curvature, and convexity

Curvature tells us how the slope changes.

The squared MHP error is a bowl with one bottom.
Pause, type, break, rebuild.

Segment 8 • Taylor expansion

Zooming in turns a smooth function into a local polynomial.

$$\frac{1}{2} \quad 2$$

A local cost approximation is useful only near the appraisal point.
Pause, type, break, rebuild.

Segment 9 • Partial derivatives

Freeze every dial except the one being nudged.

Cable length and terrain index are separate directions on an error surface.

Pause, type, break, rebuild.

Segment 10 • The gradient and directional derivatives

The gradient points toward steepest ascent.

Negative gradient reduces the MHP cost error.
Pause, type, break, rebuild.

Segment 11 • Jacobian and Hessian

Jacobians describe vector outputs; Hessians describe curvature.

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A positive-semidefinite MHP Hessian means convexity in every direction.

Pause, type, break, rebuild.

Segment 12 • Multivariable chain rule and backpropagation

Backpropagation is the chain rule applied right to left.

A prediction pipeline passes MHP feature sensitivity through layers.

Pause, type, break, rebuild.

Segment 13 · Differentiating with respect to a vector

A vector gradient stacks one partial per entry of β .

T

β holds the cable and terrain coefficients.

Pause, type, break, rebuild.

Segment 14 • Derive the OLS normal equations

Setting the squared-error gradient to zero yields the normal equations.

$$T^{-1} T$$

Use the book's MHP projects and million-PKR costs.
Pause, type, break, rebuild.

Segment 15 • Critical points and the first-order condition

A zero gradient is a candidate, not automatically a minimum.

Use curvature to classify an MHP fitting solution.

Pause, type, break, rebuild.

Segment 16 · Gradient descent, built by hand

Repeated small negative-gradient steps reduce error.

The MHP toy loop shrinks error at $\eta=.01$ and explodes at $\eta=5$.
Pause, type, break, rebuild.

Segment 17 · Batch, stochastic, and mini-batch variants

The gradient can use all, one, or a small batch of projects.

The data unit remains an MHP project.
Pause, type, break, rebuild.

Segment 18 • Constrained optimisation and Lagrange multipliers

At a constrained optimum, objective and constraint gradients align.

Ridge and lasso are constrained least-squares ideas.
Pause, type, break, rebuild.

Segment 19 • Finite differences and gradient checking

Numerical derivatives are a debugging instrument.

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Check an MHP gradient before trusting a training loop.
Pause, type, break, rebuild.

Segment 20 • Symbolic differentiation with SymPy

Symbolic tools check reasoning; they do not replace it.

Confirm the symbolic MHP squared-error gradient.

Pause, type, break, rebuild.

Segment 21 · A tiny autodiff engine from scratch

A computation graph stores local derivatives and walks backward.

Re-solve the MHP toy error without an autodiff library.
Pause, type, break, rebuild.

Segment 22 • Integrals as accumulated area

Integration accumulates; differentiation gives the local rate.

Later survival and probability chapters use this accumulation.
Pause, type, break, rebuild.

Segment 23 • Reading a traceback

The last line translates a failure into plain English.

Use shape, key, and type failures with MHP records.
Pause, type, break, rebuild.

Segment 24 · Rubber duck and the Day 0E exit check

Explain each line, then prove the calculus with a capstone.

Return to MHP projects, cable length, terrain, and cost.
Pause, type, break, rebuild.

Exit check

Compute a derivative numerically; explain a partial derivative; explain gradient descent; translate a `KeyError`, `TypeError`, and shape mismatch.

Speaker notes